Semantic Benchmark for PDF/UA

Introduction

This document is intended as a small semantic benchmark for PDF/UA generation and structure recovery. It contains headings, paragraphs, lists, links, code blocks, a blockquote, a table, a figure, and footnotes.

The main goal is to test whether common authoring pipelines preserve document structure. For a general reference, see the Markdown Guide.

Headings and Paragraphs

A structured document should preserve heading hierarchy. This paragraph contains **bold text**, *italic text*, and `inline code`.

Inline Semantics

Semantic preservation matters because a heading is not just larger text. A machine-readable document should retain these distinctions[1].

Lists

Below is an unordered list with nesting:

- First bullet item
- Second bullet item
  - Nested bullet item
  - Link to W3C

Below is an ordered list with nesting:

1. Prepare the source file.
2. Export the document to PDF.
   1. Enable accessibility tags if the tool supports them.
   2. Record the export settings.
3. Validate the resulting PDF.
4. Compare the extracted output against the source.

Blockquote

> A document can look visually correct while still being structurally broken. Accessibility and machine readability depend on logical structure.

Code Block

The following shell command illustrates a simple conversion workflow:

```
pandoc doc-a-semantic-article.md -o doc-a.tex
```

```
lualatex doc-a.tex
```

A second code example in Python:

```
from pathlib import Path

source = Path("doc-a-semantic-article.md")
print(f"Processing {source.name}")
```

Table Example

Element	Example	Purpose
Heading	'# Introduction'	Encodes section hierarchy
Emphasis	'*'	Marks strong importance
Link	'[]()'	Preserves navigation and references
Footnote	'^1'	Adds supporting notes
Code	'inline'	Distinguishes literal code from text

Figure Example



Figure 1: Figure A1. Example figure used to test image handling and alt text.

Footnotes and Closing

This sentence includes a footnote for testing purposes.[2] This second sentence includes another footnote to test multiple notes in the same document.[3]

In summary, this document is meant to be simple, readable, and rich enough to test whether structure survives across authoring, PDF generation, validation, and extraction workflows.

[1] In this benchmark, "structure" includes headings, lists, tables, figures, links, reading order, and metadata.

[2] A footnote should remain associated with the correct reference point in the main text.

[3] Some pipelines preserve footnote text but lose the reference structure or numbering semantics.